

Lingjun Meng

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Education

Imperial College London (Business School) Sep. 2023 – Now

PhD in Analytics & Operations (Fully funded by the [Business School GTA Scholarship](#))

Advisors: [Prof. Wolfram Wiesemann](#), [Prof. Ryan Cory-Wright](#)

Swiss Federal Institute of Technology in Lausanne (EPFL) Sep. 2021 – Aug. 2023

M. S. in Mechanical Engineering GPA: 5.51/6.0 (equivalent to UK distinction)

Funding: [EPFL Excellence Fellowship](#)

Xi'an Jiaotong University (XJTU) Sep. 2016 – June 2020

B. E. in Energy Engineering GPA: 92.65/100 (Ranking: 2/18)

University of Minnesota (UMN) June 2018 – Aug. 2018

Summer Exchange GPA: 3.89/4.0

Research Interest

Methodologies: optimization theory, stochastic process

Topics: Data-driven decision making under uncertainty, revenue management, production and operations management

Working Paper

L. Meng, R. Cory-Wright, and W. Wiesemann (2024). A Scalable Approximation Algorithm for Distributionally Robust Optimization.

Publication

L. Meng, J. Coldenhoff, P. Kendrick, T. Stojkovic, A. Harper, K. Ratmanski, M. Cernak. (2023). On real-time multi-stage speech enhancement systems". *ICASSP 2024* [[paper](#)] [[website](#)]

Teaching

Math and Statistics Foundation for Analytics (MSc Business Analytics) Imperial College London

Teaching Assistant 2024 Fall

Internship

Two-stage learning for real-time speech signal separation and enhancement Feb. 2023 – Aug. 2023

Research Intern, Full-time, Advisor: *Dr. Milos Cernak*

CTO office, Logitech Europe S.A., Switzerland

- Enhance the speech quality corrupted by additive noise and nonlinear distortions based on two-stage learning.
- Propose a lightweight two-stage network which consists of a Mel-scale magnitude masking model in the first stage and a complex spectrum mapping model in the second stage. The proposed two-stage network with optimal training scheme could achieve a similar performance to a four-times larger open-source model.

Research Project

A Scalable Approximation Algorithm for Distributionally Robust Optimization Sep. 2023 – Now

Research Assistant, Advisor: *Prof. Wolfram Wiesemann*, *Prof. Ryan Cory-Wright*

Imperial Business School, United Kingdom

- Design an efficient and tractable approximation scheme for Wasserstein DRO problems that leverages only two extreme solutions, the solution of nominal stochastic problems and high ambiguity problems.
- Provide a rigorous error bound for the approximation scheme and study the approach under various practical applications including portfolio selection, facility location and energy system operation.

Online bandit algorithms based on distributionally robust optimization

Feb. 2022 – Oct. 2022

Research Assistant, Advisor: *Prof. Daniel Kuhn*

Risk Analytics and Optimization Chair, EPFL, Switzerland

- Develop bandit algorithms based on DRO for online decision-making under uncertainty. Reformulate the distributionally robust bandit algorithms to tractable convex program.
- Derive non-asymptotic regret upper bounds for the policies based on concentration theorems and mathematical analysis techniques, which give a theoretical convergence guarantee for the developed DRO bandit algorithms.

Understanding test-time adaptation through synthetic simulation

Sep. 2021 – Jan. 2022

Research Assistant, Advisor: *Prof. Alexandre Alahi*

Lab of Visual Intelligence for Transportation, EPFL, Switzerland [[Project Repo](#)]

- Implement several test-time adaptation methods (TTT, TTT++, TENT, SHOT) for improving the machine learning robustness to distribution shifts.
- Design synthetic experiments to control domain shifts. Compare and analyze the mechanism of different methods under various domain shifts.